National Neighborhood Data Archive (NaNDA): Law Enforcement Organizations by Census Tract, United States, 2003-2017

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Overview and Data Dictionary

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Dataset Overview

Description

This dataset contains measures of the number and per capita density of law enforcement and safety organizations—such as police and fire departments, courts, jails, and lawyers—per United States census tract from 2003 through 2017.

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Data Sources

The North American Industry Classification System (NAICS) was used to identify law enforcement organizations. NAICS was developed by the United States Office of Management and Budget and is “the standard used by Federal statistical agencies in classifying business establishments for the purpose of collecting, analyzing, and publishing statistical data related to the U.S. business economy” (United States Census Bureau, 2017).

Establishment data was taken from the National Establishment Time Series (NETS) database, which provides annual records of the US economy for more than 60 million private for-profit and nonprofit establishments and government agencies (Walls & Associates, 2017). This longitudinal data source classifies establishments by NAICS code and provides detailed business microdata and address history. NETS is produced in most years by Walls and Associates using archival establishment data from Dun & Bradstreet. While it currently includes data from 1990 to 2015 and 2017, only data from 2003-2015 and 2017 were used to create this dataset. (NETS data was not produced for 2016.)

Census tracts for establishments were determined using the 2010 TIGER/Line shapefiles produced by the United States Census Bureau (United States Census Bureau, 2010).

Census tract population figures for 2000-2010 were taken from Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019). Population figures for 2011-2017 were obtained from Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020).

Coverage

The dataset contains one observation per census tract per year in the United States, inclding Puerto Rico and the US Virgin Islands. Other US island territories are excluded.

Methodology

This dataset was created to contribute to the discussion about many possible meanings and implications of the spatial distribution of law enforcement and public safety organizations, including police and fire departments, jails, lawyers and prosecutors, and courts. From a public safety perspective, the location of the nearest police or fire station impacts response times during emergency situations (Murray, 2013, Stassen & Ceccato, 2019). Some researchers have examined the role of neighborhood justice centers and courts in reducing arrests and recidivism (Kilmer & Sussel, 2014, Murray, 2009), as well as the role of neighborhood lawyers in advancing community improvements and social justice (Alvarez et al. 2017). The negative health effects of police presence in Black neighborhoods (Sewell et al., 2020) is also an area of focus.

Our research team identified the following NAICS codes as representing types of law enforcement organizations:

- 922: all law enforcement organizations (referred to in NAICS as “justice, public order, and safety activities”).
- 922110: courts.
- 922120: police departments.
- 922130: lawyers and attorneys, specifically those that provide legal services for the government, such as prosecutors and public defenders.
- 541110: other lawyers.
- 922140: prisons, jails, and other correctional facilities.
- 922160: fire departments.
- 922190: other organizations supporting law enforcement agencies, such as government emergency planning offices, public safety bureaus, and consumer product safety commissions.

One additional type of law enforcement organization, parole and probation officers (922150), was omitted because no results were found in the NETS database.

Note that NAICS codes are hierarchical, and three-digit codes are inclusive of all six-digit codes that start with the same pattern. For example, if a tract contains two law enforcement organizations (922), one police station (922120), and one jail (922140), the total number of law enforcement organizations is two, not four.

For each NAICS code we identified, we queried the NETS database for all establishments that were open at some time between 2003 and 2015 and in 2017 (NETS does not contain data for 2016), and their location during each year. Note that NETS data were not cleaned or verified during this process, and that NETS data has limitations and inaccuracies as described in Gomez-Lopez et al. (2017) and Finlay et al. (2019).

We determined each establishment’s census tract for each year by mapping its latitude and longitude provided in the NETS database to the 2010 version of the TIGER/Line shapefiles from the US Census Bureau.

Using these locations, we created three variables to represent establishment counts per tract. The first count variable is the total of all establishments with that NAICS code in the tract. The second is the count of just establishments with non-zero sales figures in the current year, which can be used where desired to filter out establishments with very low business volume. The third is the count of establishments employing more than one person, which can be used to filter out very small owner-operated businesses if desired. The most useful variable for a given research question may vary by establishment type, so we have created three count types for all NAICS codes so that users may make their own decision about the version that best meets their needs.

We then calculated population densities (per 1000 people) for each of the three count variables for each NAICS code. Tract populations for 2003-2009 come from the NaNDA dataset [Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010](). Values for each year are interpolated. For 2010-2012 and 2013-2017, we used the tract population from the NaNDA dataset [Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017](). Population figures in this dataset were taken from the 2012 and 2017 ACS five-year estimates, respectively, in order to avoid overlapping ACS estimates as advised by the Census Bureau, and for compatibility with other NaNDA datasets. Population figures for 2010-2017 are not interpolated. Consult the codebooks for the NaNDA datasets Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019) and Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020) for details on why these population figures were selected and how 2003-2009 population was interpolated.

We also created three variables representing the density of establishments per square mile. The area is the land area of the tract from the 2010 version of the TIGER/Line shapefiles. As a result, all area densities are standardized to 2010 census geography and are comparable across years across the entire dataset.

Population density was set to -1 for tracts with zero population in a given year. Population and area densities are missing for tracts in Puerto Rico and the US Virgin Islands, which are not included in the TIGER/Line shapefiles. Population and area densities are also missing in certain years for a few census tracts whose FIPS codes changed at some point between 2003 and 2017.

Finally, we merged counts and densities of all selected NAICS codes to create an aggregate dataset containing tract population, area, and count and density variables for all law enforcement organizations per tract in each year.

Usage Note

Data users interested in the spatial distribution of lawyers' offices will note that they are represented by two NAICS codes: lawyers providing state-related counsel, e.g. those prosecuting crimes or defending the accused (922130) and lawyers or law firms providing legal counsel on any legal matters, including criminal, corporate, estate, and tax law (541110). Different types of lawyers in a neighborhood may or may not be relevant to the research question being asked. We have therefore included both NAICS codes so that users may choose which types of lawyers to include in their analysis.

Variables

For additional information on the definition of each NAICS code, see the Methodology section of this document and the 2017 NAICS website: https://www.census.gov/cgi-bin/sssd/naics/naicsrch?chart=2017.

Variable	Type	Obs	Unique	Mean	Min	Max	Label
tract_fips10	string	1036462	74033	.	.	.	Census tract FIPS code (2010 TIGER/Line shapefiles)
year	int	1036462	14	2009.571	2003	2017	Year
population	float	1020881	250271	4208.894	0	65528	Tract population
aland10	float	1022434	72534	48.07265	0	85425.73	Tract land area, square miles
count_922	int	1036462	93	1.174898	0	105	# all law enforcement orgs
count_sales_922	int	1036462	17	0.0600215	0	35	# all law enforcement orgs w/ sales>0
count_emp_922	int	1036462	90	1.090885	0	97	# all law enforcement orgs w/ 2+ employees
popden_922	float	1015655	179073	1.282076	-1	289507.3	# all law enforcement orgs per 1000 people
popden_sales_922	float	1015107	28029	0.0257507	-1	2646.964	# all law enforcement orgs per 1000 people w/ sales>0
popden_emp_922	float	1015640	172663	1.24912	-1	289507.3	# all law enforcement orgs per 1000 people w/ 2+ employees
aden_922	float	1017996	85839	0.7869053	0	491.838	# all law enforcement orgs per sq mile
aden_sales_922	float	1017996	16283	0.0627052	0	88.20026	# all law enforcement orgs per sq mile w/ sales>0
aden_emp_922	float	1017996	84874	0.7492423	0	491.838	# all law enforcement orgs per sq mile w/ 2+ employees
count_922110	int	1036462	52	0.2242002	0	57	# courts
count_sales_922110	int	1036462	7	0.0157777	0	6	# courts w/ sales>0
count_emp_922110	int	1036462	49	0.2118235	0	56	# courts w/ 2+ employees
popden_922110	float	1015128	52546	0.0774658	-1	333.3333	# courts per 1000 people
popden_sales_922110	float	1015053	8583	0.0047329	-1	128.7576	# courts per 1000 people w/ sales>0
popden_emp_922110	float	1015121	50950	0.0728746	-1	333.3333	# courts per 1000 people w/ 2+ employees
aden_922110	float	1017996	23069	0.2230316	0	231.5854	# courts per sq mile
aden_sales_922110	float	1017996	7309	0.0216923	0	52.7514	# courts per sq mile w/ sales>0
aden_emp_922110	float	1017996	22883	0.214785	0	231.5854	# courts per sq mile w/ 2+ employees
count_922120	int	1036462	23	0.2551517	0	23	# police stations

Variable	Type	Obs	Unique	Mean	Min	Max	Label
count_sales_922120	int	1036462	6	0.0090587	0	5	# police stations w/ sales>0
count_emp_922120	int	1036462	23	0.2433751	0	23	# police stations w/ 2+ employees
popden_922120	float	1015341	75017	0.1594268	-1	10277.78	# police stations per 1000 people
popden_sales_922120	float	1015058	5384	0.0042053	-1	333.3333	# police stations per 1000 people w/ sales>0
popden_emp_922120	float	1015340	73279	0.1546963	-1	10277.78	# police stations per 1000 people w/ 2+ employees
aden_922120	float	1017996	23628	0.1956017	0	139.4676	# police stations per sq mile
aden_sales_922120	float	1017996	2051	0.0087834	0	21.30508	# police stations per sq mile w/ sales>0
aden_emp_922120	float	1017996	23389	0.1888626	0	139.4676	# police stations per sq mile w/ 2+ employees
count_922130	int	1036462	52	0.0592622	0	73	# prosecutors/public defenders
count_sales_922130	int	1036462	12	0.0062617	0	31	# prosecutors/public defenders w/ sales>0
count_emp_922130	int	1036462	48	0.0553865	0	64	# prosecutors/public defenders w/ 2+ employees
popden_922130	float	1015100	19993	0.0365908	-1	2646.964	# prosecutors/public defenders per 1000 people
popden_sales_922130	float	1015055	4600	0.0070946	-1	2646.964	# prosecutors/public defenders per 1000 people w/ sales>0
popden_emp_922130	float	1015100	18910	0.0321681	-1	2646.964	# prosecutors/public defenders per 1000 people w/ 2+ employees
aden_922130	float	1017996	5778	0.0844382	0	285.5881	# prosecutors/public defenders per sq mile
aden_sales_922130	float	1017996	1785	0.0094392	0	53.69799	# prosecutors/public defenders per sq mile w/ sales>0
aden_emp_922130	float	1017996	5556	0.0796323	0	285.5881	# prosecutors/public defenders per sq mile w/ 2+ employees
count_541110	int	1036462	878	5.503286	0	5169	# lawyers
count_sales_541110	int	1036462	825	5.101438	0	4764	# lawyers w/ sales>0
count_emp_541110	int	1036462	803	4.812795	0	4753	# lawyers w/ 2+ employees
popden_541110	float	1015611	306777	32.72322	-1	1.17E+07	# lawyers per 1000 people
popden_sales_541110	float	1015583	297722	32.49593	-1	1.17E+07	# lawyers per 1000 people w/ sales>0
popden_emp_541110	float	1015590	285835	32.35035	-1	1.17E+07	# lawyers per 1000 people w/ 2+ employees
aden_541110	float	1017996	219920	10.8615	0	12392.17	# lawyers per sq mile
aden_sales_541110	float	1017996	214156	10.08877	0	11675.76	# lawyers per sq mile w/ sales>0
aden_emp_541110	float	1017996	213091	9.44511	0	11829.27	# lawyers per sq mile w/ 2+ employees
count_922140	int	1036462	22	0.0798129	0	22	# jails/prisons
count_sales_922140	int	1036462	6	0.0075044	0	10	# jails/prisons w/ sales>0
count_emp_922140	int	1036462	21	0.0764379	0	22	# jails/prisons w/ 2+ employees
popden_922140	float	1015094	28901	0.0289629	-1	1729.786	# jails/prisons per 1000 people

Variable	Type	Obs	Unique	Mean	Min	Max	Label
popden_sales_922140	float	1015052	4814	0.0020763	-1	4.926108	# jails/prisons per 1000 people w/ sales>0
popden_emp_922140	float	1015094	28213	0.0279997	-1	1729.786	# jails/prisons per 1000 people w/ 2+ employees
aden_922140	float	1017996	7444	0.0614386	0	248.873	# jails/prisons per sq mile
aden_sales_922140	float	1017996	1503	0.0067152	0	28.29135	# jails/prisons per sq mile w/ sales>0
aden_emp_922140	float	1017996	7324	0.0588825	0	232.2815	# jails/prisons per sq mile w/ 2+ employees
count_922160	int	1036462	17	0.5153792	0	17	# fire departments
count_sales_922160	int	1036462	6	0.0171014	0	5	# fire departments w/ sales>0
count_emp_922160	int	1036462	15	0.4675782	0	17	# fire departments w/ 2+ employees
popden_922160	float	1015338	122478	0.9428187	-1	289507.3	# fire departments per 1000 people
popden_sales_922160	float	1015062	10457	0.0056507	-1	86.43819	# fire departments per 1000 people w/ sales>0
popden_emp_922160	float	1015329	115905	0.9272894	-1	289507.3	# fire departments per 1000 people w/ 2+ employees
aden_922160	float	1017996	44805	0.1855941	0	76.84969	# fire departments per sq mile
aden_sales_922160	float	1017996	4028	0.0112006	0	37.15375	# fire departments per sq mile w/ sales>0
aden_emp_922160	float	1017996	44009	0.1734499	0	76.84969	# fire departments per sq mile w/ 2+ employees
count_922190	int	1036462	15	0.0455212	0	15	# other law enforcement orgs
count_sales_922190	int	1036462	4	0.0043899	0	4	# other law enforcement orgs w/ sales>0
count_emp_922190	int	1036462	16	0.0405823	0	15	# other law enforcement orgs w/ 2+ employees
popden_922190	float	1015163	20440	0.0385567	-1	2508.074	# other law enforcement orgs per 1000 people
popden_sales_922190	float	1015065	3332	0.0020211	-1	186.7105	# other law enforcement orgs per 1000 people w/ sales>0
popden_emp_922190	float	1015163	18808	0.0357734	-1	2508.074	# other law enforcement orgs per 1000 people w/ 2+ employees
aden_922190	float	1017996	5347	0.0408588	0	46.10981	# other law enforcement orgs per sq mile
aden_sales_922190	float	1017996	1288	0.0049482	0	25.75386	# other law enforcement orgs per sq mile w/ sales>0
aden_emp_922190	float	1017996	5109	0.0375484	0	46.10981	# other law enforcement orgs per sq mile w/ 2+ employees

References

Alvarez, A., Bennett, S., Howells, L., & Lieberman, H. (2017). Teaching and practicing community development law: Lawyers and clients as trusted neighborhood problem solvers. *Clinical Law Review 23*(2): 577-610.

Clarke, P. J., & Melendez, R. (2019). National Neighborhood Data Archive (NaNDA): Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (V1) [Data set]. Inter-university Consortium for Political and Social Research. doi: 10.3886/E111107V1

Finlay, J. M., Esposito, M., Kim, M. H., Gomez-Lopez, I. N., & Clarke, P. J. (2019). Closure of "Third Places"? Exploring Potential Consequences for Collective Health and Wellbeing. *Health & Place, 60*, 102225. doi: 10.1016/j.healthplace.2019.102225

Finlay, J. M., Li, M., Esposito, M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., & Chenoweth, M. M. (2020). National Neighborhood Data Archive (NaNDA): Grocery Stores by Census Tract, United States, 2003-2017 (V1) [Data set]. Inter-university Consortium for Political and Social Research. doi: 10.3886/E123001V1

Gomez-Lopez, I. N., Clarke, P. J., Hill, A. B., Romero, D. M., Goodspeed, R., Berrocal, V. J., Vinod Vydiswaran, V.G, & Veinot, T. C. (2017). Using Social Media to Identify Sources of Healthy Food in Urban Neighborhoods. *Journal of Urban Health, 94*(3), 429–36. doi: 10.1007/s11524-017-0154-1

Kilmer, B., & Sussell, J. (2014). *Does San Francisco's community justice center reduce criminal recidivism?* Rand Corporation. https://www.rand.org/content/dam/rand/pubs/research_reports/RR700/RR735/RAND_RR735.pdf

Melendez, R., Clarke, P. J., Khan, A. M., Gomez-Lopez, I. N., Li, M., & Chenoweth, M. (2020). National Neighborhood Data Archive (NaNDA): Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (V1) [Data set]. University of Michigan: Intra-university Consortium for Political and Social Research. doi: 10.3886/E120462V1

Murray, A. T. (2013). Optimising the spatial location of urban fire stations. *Fire Safety Journal 62*(A): 64-71. doi: 10.1016/j.firesaf.2013.03.002

Murray, S. (2009). Keeping it in the neighborhood? Neighborhood courts in the Australian context. *Monash University Law Review 35*(1), 74-95.

Sewell, A. A., Feldman, J. M., Ray, R., Gilbert, K. L., Jefferson, K. A. & Lee, H. (2020). Illness spillovers of lethal police violence: the significance of gendered marginalization. *Ethnic and Racial Studies*. DOI: 10.1080/01419870.2020.1781913

Stassen, R., & Ceccato, V. (2019). Police accessibility in Sweden: An analysis of the spatial arrangement of police services. *Policing: A Journal of Policy and Practice*, 1-16. doi: 10.1093/police/paz068

United States Census Bureau (2010). TIGER/Line Shapefiles, 2010 Census Tracts (2010 version) [Data set]. https://www2.census.gov/geo/tiger/TIGER2010/TRACT/2010/tl_2010_us_tract10.zip.

United States Census Bureau (2017). North American Industry Classification System (2017 version) [Data set]. https://www.census.gov/cgi-bin/sssd/naics/naicsrch?chart=2017

Walls & Associates (2017). National Establishment Time Series (NETS) Database.